Alexander Greif

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Profile

I work with researchers and students to address empirical questions using statistical methods, from shaping research questions to analysing data and interpreting results. My experience spans work within university research teams, independent statistical consulting across disciplines, and teaching and tutoring. My ongoing studies in mathematical-technical software development complement this work with a theoretical grounding in mathematics and computer science.

Research and Data Analysis Experience

Statistical Consultant - Self-employed — Remote
Working with direct private clients and for statistical consultancy NOVUSTAT — Apr 2023 – present

- Provide methodological and statistical advice, helping clients develop research questions and hypotheses, plan data collection and select statistical methods.
- Conduct data processing and statistical analyses in R, SPSS, and Excel, either independently or alongside clients.
- Deliver documented analysis projects, including commented code, tables, figures, and reports; guide clients through their structure and explain methods, results and limitations.
- Help clients communicate methods and findings in their academic work, advising on which results to emphasise and how to present them clearly through tables, figures and text.

Student Research Assistant - University of Münster — Münster, Germany
Education and Teaching in Sports, Institute of Sport and Exercise Sciences — Apr 2021 – Apr 2023

- Advised researchers across multiple research and doctoral projects on methodological topics, including hypothesis formulation, selection of statistical methods and interpretation of findings.
- Conducted statistical analyses and handled data processing and management across these projects; produced reports, tables and figures, and explained methods and findings to researchers.
- Developed a structured R project integrating data preparation, statistical analysis and R Markdown reporting for a controlled intervention study with repeated measurements, combining data from multiple sources.
- Conducted statistical analyses of cross-sectional survey data and wrote the quantitative methods and results sections of the resulting co-authored article on gender differences in student satisfaction.

Sports Science Intern - Plymouth Argyle FC Academy — Plymouth, United Kingdom
Erasmus-supported university placement — Aug 2019 – Dec 2019

- Analysed GPS and heart-rate data in Excel to track changes in player workload and develop physical demand profiles for training drills, supporting the evaluation and planning of training programmes.
- Supported data collection for player monitoring through GPS and heart-rate measurements, fitness testing, wellness questionnaires and anthropometric assessments of growth and maturation.
- Led on-field and gym-based strength and conditioning sessions for academy players, as well as individualised training for injury recovery and load management, in coordination with coaches and physiotherapists.

Publications

Magner, A., **Greif, A.**, & Kohake, K. (2026). Studienzufriedenheit im Sportlehramtsstudium: Eine Mixed-Methods-Studie unter besonderer Berücksichtigung des Geschlechts [Student satisfaction in physical education teacher education: A mixed-methods study with particular attention to gender]. *Beiträge zur Hochschulforschung*, *48*(2), 110–139. https://doi.org/10.58069/bzh.2026.2.192

Technical Skills

R, Python, SQL, SPSS, Excel, R Markdown, Git/GitHub

Education

FernUniversität in Hagen — Germany
Mathematical-Technical Software Development - B.Sc. programme, ongoing — Oct 2022 – present
Part-time distance study; 160 ECTS completed as of July 2026

- Mathematical and statistical foundations covering probability theory, measure and integration theory, parametric statistics, linear algebra, real analysis and numerical methods.
- Computer science and software development covering data structures and algorithms, object-oriented programming, relational databases and software engineering.

University of Münster — Münster, Germany
Sports, Exercise and Human Performance - M.Sc. programme, discontinued — Oct 2020 – Mar 2023
Discontinued without completing the degree

- Collected and analysed biomechanical and neurophysiological data, including kinematic and kinetic measurements, EMG and EEG; applied computational models to estimate joint moments and internal forces.

University of Münster — Münster, Germany
Human Movement in Sports and Exercise - B.Sc. — Oct 2017 – Mar 2021
Final grade: 1.8 (German grading scale)

- Coursework covering biomechanics, motor control, exercise physiology, sport psychology and social aspects of sport.
- Training in research design, statistical methods, critical evaluation of scientific evidence and scientific writing; practical experience in planning and conducting empirical studies and communicating findings.

Selected Research and Academic Projects

The integration of external and internal match load with contextual factors to assess training status in football — Mar 2021
Bachelor's thesis, University of Münster; thesis grade: 1.3 (German grading scale)

- Compared multilevel regression models of mean match heart rate in R, accounting for repeated observations within players; likelihood-ratio tests showed that adding weather and match location improved fit over a model based on GPS-derived movement measures alone.
- Used multilevel models to evaluate deviations between observed and predicted heart rate as a potential fatigue indicator, testing their association with closely scheduled matches where fatigue was expected; found no clear association.

Survival Analysis — May 2025
Statistical practical assignment in Mathematical Statistics, FernUniversität in Hagen

- Analysed time-to-death data with right-censored observations in R, using Kaplan–Meier curves and a log-rank test to compare treatment and placebo groups.
- Conducted an event-based power analysis to assess statistical power and required event counts.
- Compared age distributions between groups using the Wilcoxon rank-sum test to assess potential confounding of the treatment comparison.

Cyclistic Case Study — Sept 2022
Capstone project, Google Data Analytics Professional Certificate

- Processed and cleaned 5.7 million bike-share trip records in R, documenting decisions and their rationale.
- Analysed temporal and geographic usage patterns of casual riders and annual members, examining differences in trip duration, timing and station use.
- Produced R Markdown reports, visualisations and a presentation communicating findings and recommendations for converting casual riders into annual members.

Teaching Experience

Freelance IT Instructor - Qualidy
Remote
Teaching in a vocational retraining programme at Berufliche Fortbildungszentren der Bayerischen Wirtschaft (bfz)
Apr 2026 – present

- Plan and deliver one-week online modules for adults retraining as IT specialists, covering programming, databases, and Linux and Windows system administration.
- Develop teaching materials, practical exercises and assessments, and set up virtual lab environments for hands-on learning.

Tutor in Statistics, Mathematics and Computer Science - Self-employed
Remote
Direct private clients; previously also for Studienkreis
Apr 2023 – present

- Provide individual tutoring for university students in statistics, applied mathematics, programming and databases, supporting exam preparation and ongoing coursework.

Data Analysis Tutor - DataCraft
Remote
Working student position
Mar 2024 – May 2025

- Provided daily tutoring throughout a one-year data analyst training programme, helping participants acquire, process, analyse and visualise data and communicate findings using tools such as Python, SQL and Power BI.
- Guided participants through individual and group assignments and practical projects, fostering critical thinking and independent problem-solving by prioritising hints and targeted questions over direct answers.
- Reviewed, improved and developed learning materials.

Languages

German: Native

English: C1 (CEFR, self-assessed)

Spanish: B1 (CEFR, self-assessed)

Professional Development and Certifications

Google Data Analytics Professional Certificate
Sept 2022
Training in data preparation, analysis and visualisation using spreadsheets, SQL and R

Additional Professional Experience

Telephone Interviewer - Ipsos
Amsterdam, Netherlands
Part-time market research
Aug 2014 – June 2015

- Conducted telephone interviews with individuals and businesses for market research studies.